

Assignment 10

ROLL NO:CSB24083

Name:Dipankar Bezbaruah

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class Address:
    def __init__(self, street, city, zipCode):
        self.street = street
        self.city = city
        self.zipCode = zipCode

    def __str__(self):
        return f"{self.street}, {self.city} - {self.zipCode}"

class Student:
    def __init__(self, name, age, address, courses=None):
        self.name = name
        self._age = None
        self.age = age
        self.address = address
        self.courses = courses if courses is not None else []

    @property
    def age(self):
        return self._age

    @age.setter
    def age(self, value):
        if not isinstance(value, int) or value <= 0 or value >
120:
            raise ValueError("Age must be a valid positive
integer (1-120).")
```

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        self._age = value

    def add_course(self, course):
        self.courses.append(course)

    def display(self):
        print(f"Name: {self.name}")
        print(f"Age: {self.age}")
        print(f"Address: {self.address}")
        print(f"Courses: {'', '.join(self.courses) if self.courses
else 'None'}")

class ScholarshipStudent(Student):
    def __init__(self, name, age, address, courses=None,
scholarshipAmount=0):
        super().__init__(name, age, address, courses)
        self.scholarshipAmount = scholarshipAmount

    def display(self):
        super().display()
        print(f"Scholarship Amount: ₹{self.scholarshipAmount}")

if __name__ == "__main__":

    addr = Address("MG Road", "Guwahati", "781001")

    s1 = Student("Rahul", 20, addr)
    s1.add_course("Math")
    s1.add_course("Physics")

    print("---- Student ----")
    s1.display()

    s2 = ScholarshipStudent("Anita", 22, addr, ["Biology"],
50000)

```

```
print("\n---- Scholarship Student ----")  
s2.display()
```

OUTPUT

---- Student ----

Name: Rahul

Age: 20

Address: MG Road, Guwahati - 781001

Courses: Math, Physics

---- Scholarship Student ----

Name: Anita

Age: 22

Address: MG Road, Guwahati - 781001

Courses: Biology

Scholarship Amount: ₹50000